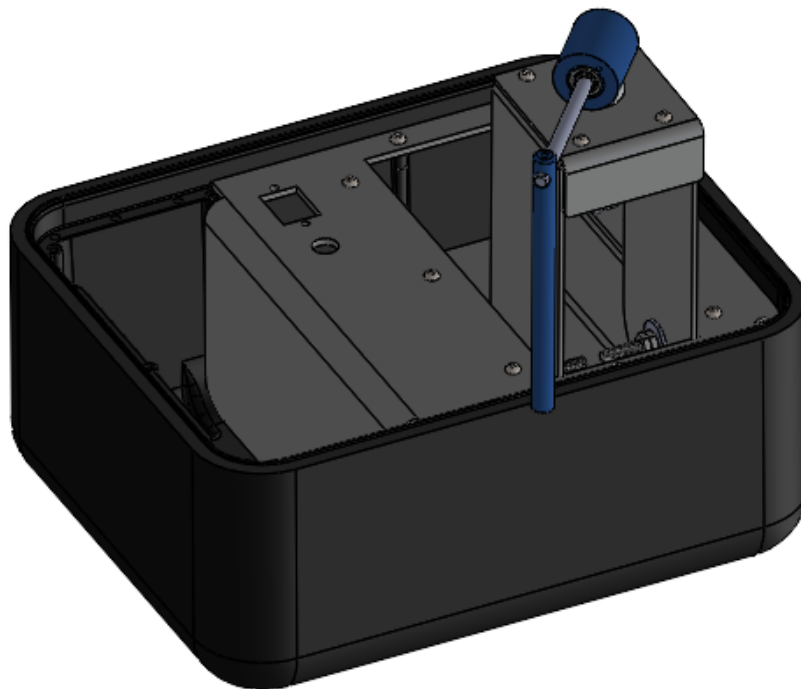


Open-Source Teaching Platform for Control Systems: Technical Instructions

Isaiah Knox, Alyssa Jackson, and Eric Pollert



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1 Introduction

1.1 Scope

This device is a tabletop electromechanical platform developed to support instruction in control systems classes. It is intended for use in undergraduate laboratories where engineering students design, implement, and evaluate feedback control strategies such as PID control on a physical system rather than in simulation alone.

The system is designed to replicate the functional behavior and key performance characteristics of the Quanser Qube-Servo 2 so that existing laboratory exercises and workflows can be maintained without modification. Core components, including the DC motor and optical encoders, are used to create a response in the motor based on feedback from the encoders. This allows the device to serve as a drop-in replacement for the Qube-Servo 2 in control systems classes, as it consists of the same major components and is capable of completing the same tasks.

This device was created as an open-source alternative to the Quanser Qube-Servo 2. All design schematics, code, and documentation are available on GitHub to anyone interested in utilizing this device. This approach is intended to prevent obsolescence tied to proprietary software or hardware dependencies and to allow future updates as course requirements or supporting technologies evolve.

These instructions describe the configuration, assembly, and operation of the device, along with the information required to reproduce and validate its performance.

1.2 Safety

- Do not expose the interior of the device to water
- Unplug the device when not in use
- Do not leave the device unattended while operating
- Keep hands and other objects away from the spinning motor and attachments while in operation.

2 Package Contents

All of the components required to make the device are listed in the tables below. The tables are broken up into the individual sub-assemblies that make up the device.

Attachment Storage

Part Name	Quantity	Supplier	Part Number / Description
Foam Panel	1	Custom	Foam Divider
Front Panel	1	Custom	Front Panel of the lid
Rear Lid Panel	1	Custom	Back Panel of the lid
6 1/2 Thread Forming Screw	4	McMaster Carr	90385A148

Box Internal Structure

Part Name	Quantity	Supplier	Part Number / Description
Plate Bend Inlet	1	Custom	Top Panel of the box interior
Housing	1	Custom	Trough for motor housing
Arduino Uno	1	Arduino Store	Arduino Uno R3
PCB	1	Custom	Circuit board that connects all electrical components
Motor Controller	1	Adafruit	DRV8871
Power Port	1	Digikey	Switchcraft 712AFMS
USB Port	1	Digikey	Cliff CP30608NX1
6-32 1/2 Machine Screw	6	McMaster-Carr	90279A148
6-32 Locknut	6	McMaste-rCarr	90631A007
4-40 1/2 Machine Screw	2	McMaster-Carr	90279A110
4-40 Nut	2	McMaster-Carr	91841A005
PCB Mounting Plate	1	Custom	Acrylic Mounting Plate for PCB
6 1/2 Thread Forming Screw	5	McMaster-Carr	90385A148
Nanuk 905 Case	1	Nanuk	Case the device is built in
Cable Gromet	1	Digikey	Essentra Components DG-10134
Power Source	1	Digikey	Mean Well GST90A24-P1M
USB A to USB B	1	Amazon	Belkin F3U133-06INCH
Encoder Cord	1	US Digital	CA-MIC4-W4-NC

Attachments

Part Name	Quantity	Supplier	Part Number / Description
Intertia Disk	1	Custom	Aluminum disk
Neodinium Magnets	8	Tinycircuits	6 mm X 4 mm
Pendulum Housing	1	Custom	Cylindrical Component of the Pendulum
Encoder	1	US Digital	E8T
M2.5x0.45 Screw	2	McMaster-Carr	92000A105
Bearings	2	McMaster-Carr	57155K323
8-32 Socket Head Screw	1	McMaster-Carr	91274A048
Pendulum Shaft	1	Custom	N/A

1/4 in Retaining Ring	2	McMaster-Carr	98410A113
4-40 1/8 in Set Screw	1	McMaster-Carr	94355A130
Pendulum Arm	1	Custom	N/A
Aux Cord	1	Amazon	3.5mm Male Plug to 4Pin Bare Wire
Encoder Cord	1	US Digital	CA-MIC4-W4-NC

Motor Housing

Part Name	Quantity	Supplier	Part Number / Description
Motor Plate	1	Custom	Top Plate of Motor Housing
Motor Legs	2	Custom	Sidewalls of Motor Housing
Motor	1	Maxon	DC-max 26 S Ø26 mm, CLL pre-cious metal brushes, sintered bearings
Encoder	1	US Digital	E8T
Adhesive	1	Amazon	3M 300lse
Aux Port	1	Digikey	Same Sky SJ1-43502PM
Pendulum Stop	1	Custom	3D printed part that protects the pendulum AUX cord
6-32 1/2 Machine	4	McMaster-Carr	90279A148
6-32 Locknut	4	McMaster-Carr	90631A007
4-40 1/2 Machine Screw	2	McMaster-Carr	90279A110
4-40 Nut	2	McMaster-Carr	91841A005
M2 8 machine Screw	2	McMaster-Carr	92005A029
Attachment Interface	1	Custom	Connects device attachments to the motor
2-56 3/16 Set Screw	1	McMaster-Carr	92311A076
10-32 1-1/4 Partially Threaded Socket Head	2	McMaster-Carr	90128A948
Nylon Washer	4	McMaster-Carr	90295a639
10-32 Nut	4	McMaster-Carr	91841A195
Neodinium Magnets	4	Tinycircuits	6 mm X 4 mm

3 Materials/Tools Required

3.1 Hardware

- Computer capable of running the software in Section 3.2
- Soldering Iron
- Pendulum encoder wires, motor wires, and motor encoder wires
- Heat Shrink

- Crescent Wrenches
- Philips Screwdrivers
- Allen Wrenches

3.2 Software

- MATLAB (Windows or Linux Versions)
- Simulink (Windows or Linux Versions)
- Arduino IDE (Windows or Linux Versions)
- All software downloaded from the "C.E.W.B" GitHub Software section

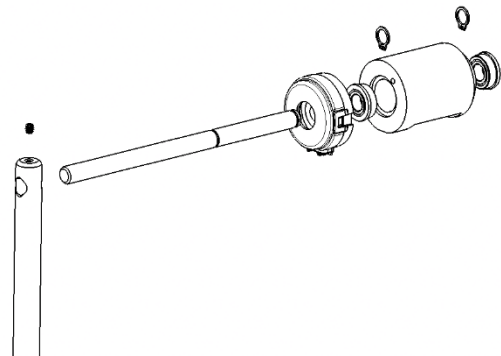
4 Procedure

4.1 Assembly

4.1.1 Pendulum Subassembly

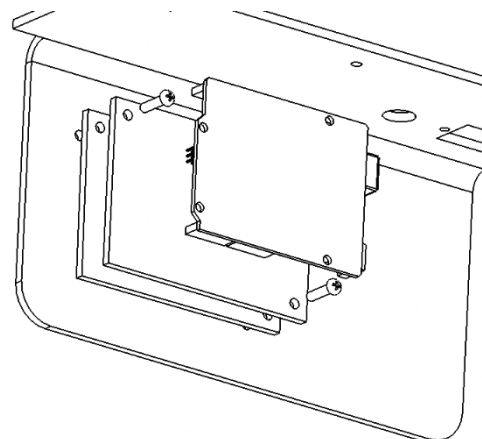
The pendulum assembly seats onto the interface on the motor via a magnetic connection. Primary parts are the pendulum mount, arm, and shaft joined with two bearings, retaining rings, and a cap screw for the arm.

- 1) Press-fit bearings into either side of the pendulum mount
- 2) Insert the pendulum shaft through both bearings in the pendulum mount
- 3) Place one retaining on the end-groove of the shaft by the butt of the mount
- 4) Place the second ring on the second groove near the face of the mount
- 5) Apply the encoder assembly over the pendulum shaft against the flat face of the mount
- 6) Insert the shaft of the assembly through the hole in the pendulum arm and secure the connection with a set screw leaving 0.25 inch of the shaft beyond the arm



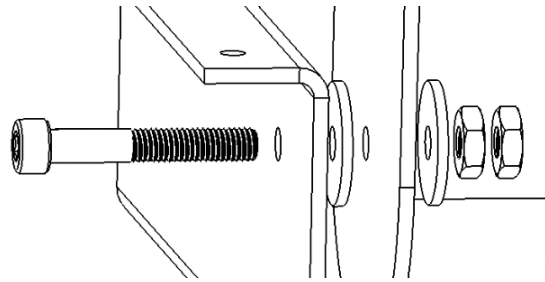
4.1.1 Internal Hardware Subassembly

- 1) Stack the PCB onto the acrylic mounting plate
- 2) Insert two screws into opposite corners of the plates and through the primary bent plate
- 3) Secure equally sized screws to the ends of each screw
- 4) Apply the Arduino to the PCB in the orientation shown



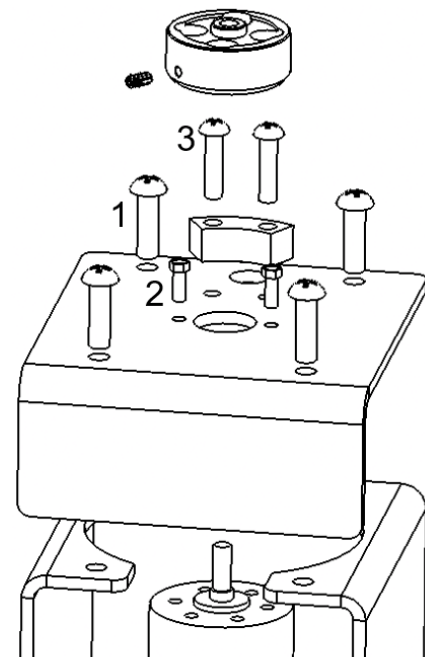
4.1.1 Hinge Subassembly

- 1) Insert the partially threaded socket head screw through the outside of the side of the trough
- 2) Apply a nylon washer to the screw against the inner face of the trough
- 3) Put one of the legs onto the screw with the bend pointing inward
- 4) Apply another nylon washer followed by two nuts to the screw
- 5) Torque the inner nut to allow movement but allow the assembly to remain upright. Apply the second nut to secure the joint. It should allow free pivot of the joint, but will remain in one place and not fall into the trough
- 6) Repeat for the second leg



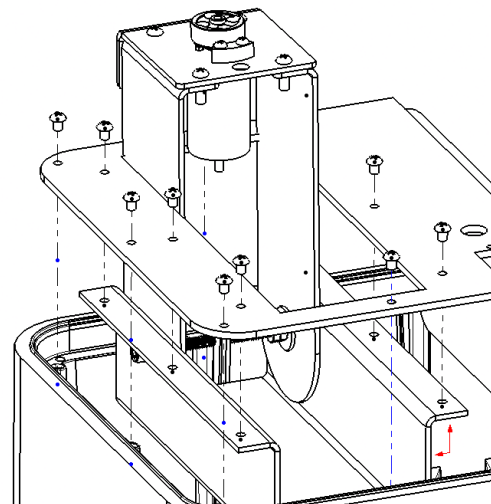
4.1.1 Motor Plate Subassembly

- 1) Secure the bent face plate to the top of the legs using the #6 screws labeled (1) in the diagram. Secure each with a #6 nut
- 2) Insert the motor through the face plate with the longer motor shaft upward. Secure the motor with two M2 screws labeled (2)
- 3) Secure the plastic stop-piece with two #4 screws labeled (3) and #4 nuts on each screw
- 4) Fit the attachment interface onto the motor shaft and secure it with a set screw



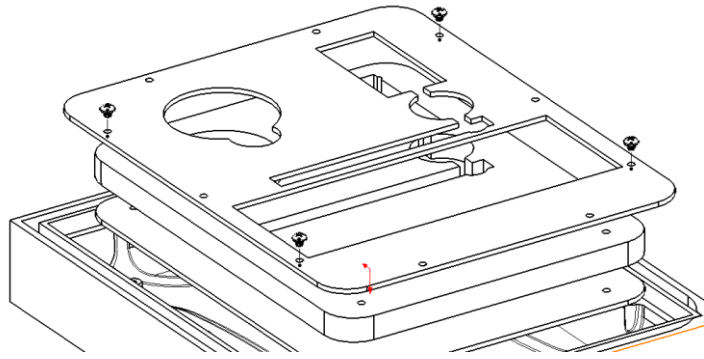
4.1.1

- 1) Place the main bent baseplate over the pendulum mount, onto the trough as shown
- 2) Secure six #6 screws with appropriate lock nuts
- 3) Secure a rubber gromet into the hole at the bottom of the trough
- 4) (Not Pictured) Thread all wires from Arduino through hole in bottom of trough
- 5) With wiring connections secured, place the full assembly into bottom of case
- 6) Secure into place with 5 thread-forming screws



4.1.1

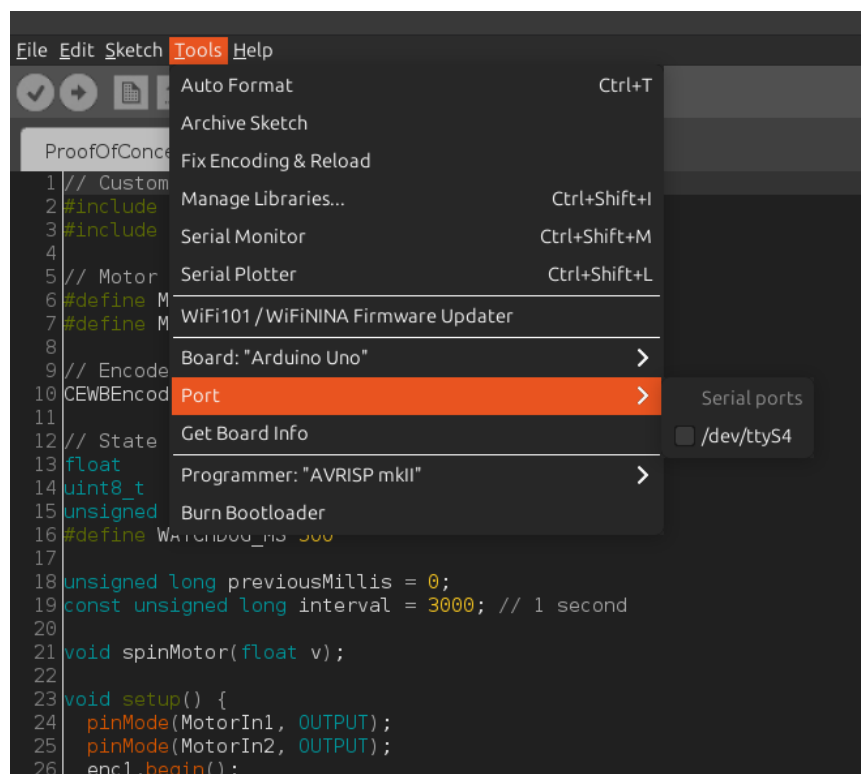
- 1) Press-fit magnets into the 8 small holes in the smaller board – Apply glue to the inner surface for each magnet individually
- 2) Glue the laser cut foam panel to the smaller, rear board. Orient the large rectangular cut on the right, with the cut for the pendulum on top
- 3) Glue the other side of the foam to the back of the larger board, aligning the cuts
- 4) After glue has cured, use thread-forming screws to secure the assembly into place



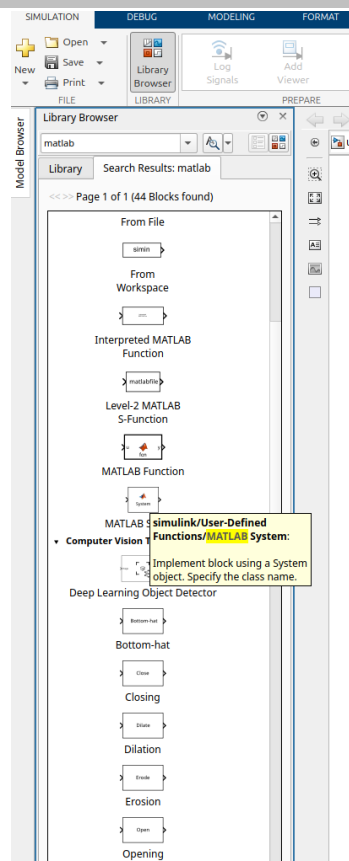
1.2 Operation

NOTE: BEFORE USE, MAKE SURE THE MOTOR HOUSING IS IN THE UPRIGHT POSITION AND THE POWER AND COMPUTER CABLES, ALONG WITH ANY OTHER OBJECTS, ARE AWAY FROM THE MOTOR SHAFT!

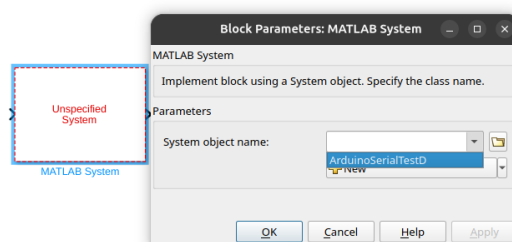
1. Create and open a blank Simulink file. This file must be created in the same folder/location as the “ArduinoSerial.m” MATLAB file for the system to work.



2. In the “ArduinoSerial.m” file, in the COMSPORT line, the port that the Arduino is connected to needs to be listed there. To check the exact name of the port, connect the Arduino to the user’s PC and open Arduino IDE. Go to *Tools*, then go to the *Ports* option. The connected port that is listed with the Arduino Uno is the port needed. Copy the port name and add it to the COMSPORT line in the “ArduinoSerial.m” file.

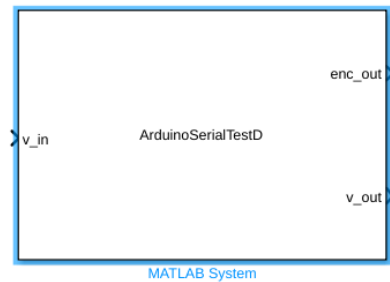


- While in Simulink, navigate to the *Library Browser* menu, find the *MATLAB System* block, and drag it onto the Simulink work area.

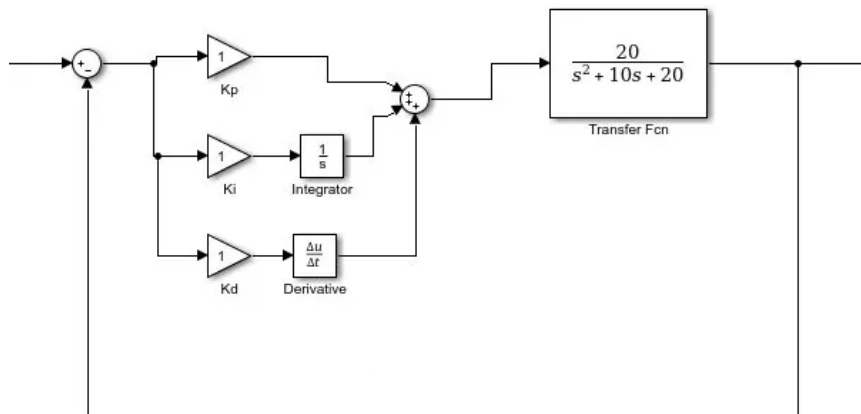


- The *MATLAB System* block will be red and unusable at first. Click on the block, then in the dropdown click on the "ArduinoSerial" option and "OK".

5. NOTE: IF THE "ARDUINOSERIAL" OPTION DOES NOT APPEAR, THAT MATLAB FILE EITHER IS NOT IN THE SAME FOLDER AS THE SIMULINK MODEL. GO BACK AND MAKE SURE IT IS THERE!



6. When the “ArduinoSerial” is selected, the *MATLAB System* will transform into a block labelled “ArduinoSerial” and have one input labeled “ v_{in} ” and two outputs labeled “ enc_out ” and “ v_out ”.



7. Once the setup is complete, the “ enc_out ” value can be used for the PID controller. The output of the “ enc_out ” is position count, so a calculation of 360 deg or $2\pi \text{ rads}$ divided by 2048 can be used to find the angle of the system, while the derivative can be but in to find the speed of the system.

1.3 Maintenance

1.4 Storage

The steps below describe how the device should be broken down for storage when not in use.

1. Unplug the USB cable from your computer and from the device, and the power cable from the device and the wall.
2. Unplug the power cable from the power supply brick and place the brick into the storage compartment on the left side of the device. Coil the power cable and place it into the storage compartment on top of the brick. Place the USB cable in the storage compartment and ensure that all of the cables are under the panel and will allow the lid to close.
3. Remove all attachments that were used and place them in their respective storage locations in the lid.

4. Fold the motor housing down in the trough, being careful to ensure that wires are not catching on anything.
5. Close the lid and lock the two latches on the side.

2 Troubleshooting

The table below provides information on how to solve common problems that might arise during use of the device. Find the problem you are experiencing under the issues column and determine the proper fix under the corrective action column.

Issue	Cause(s)	Corrective Action
Loose Motor Housing	Hinge loosening	Tighten hinge bolt and nuts
Motor Not Working	Loose/Missing Connection(s) Motor Driver Fault	Run the "MotorRamp-SpeedTest.ino" script in the "ArduinoTests" folder in "ExamplesAndTests". If it doesn't work, then open the CEWB and test connections for and make sure the CEWB is plugged in. If still not working, test/replace the motor driver
Encoder Not Working	Loose/Missing Connection(s)	Run the "Encoder-QuadTest.ino" script in the "ArduinoTests" folder in "ExamplesAndTests". If it still doesn't work, then open the encoder and ensure that it is still optical piece is on and in the right place.

3 Technical Support

For questions regarding the motor, contact maxon motors at info.us@maxongroup.com or call (508) 677-0520.

For questions regarding the optical encoders, contact US Digital at support@usdigital.com or call 800-736-0194.

The 3D models for all of the custom components can be found on the GitHub Repository by following this link

<https://github.com/aJax-EXE/C.E.W.B./tree/main/Hardware/3DModels/CurrentVersion>

For any other questions regarding the device, contact Dr. Talles Santos by email at Talles.Santos@colorado.edu

4 Spare Parts/Components

A list of all of the parts making up the device, as well as part numbers and supplier information, are listed under the package contents section.